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INVENTORY MANAGEMENT SYSTEM

EFFICIENT MOVEMENT OF GOODS AND SERVICES

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BACKGROUND

ABOUT FRACA SERVCOM...

Context: FRACA SERVCOM has grown rapidly, now selling furniture (sofas, beds, tables), electronics (TVs, fridges, audio systems), and many other product categories.

Current state: The company currently relies on manual stock tracking using spreadsheets and handwritten notes. Staff must physically check shelves or call suppliers to know stock levels.

- **Market gap within FRACA SERVCOM**: No centralized digital system to track stock across different product types (furniture vs electronics have different turnover rates). No real-time sales-purchase integration. Difficult to enforce who can adjust stock or see supplier costs.

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Problem statement

manual stock management leads to:

- Informality: FRACA SERVCOM still uses manual stock logs (spreadsheets, handwritten notes) across both furniture and electronics sections. No digital record of daily sales or purchases.
- Inefficiency: Stockouts on fast-moving electronics (e.g., phone chargers, headphones) happen frequently because reordering is not automated. Conversely, overstocking of bulky furniture (e.g., sofas) ties up warehouse space and capital.
- Trust gap: No audit trail for stock adjustments – if a sofa is damaged or an electronic item is returned, there is no system record of who approved the adjustment.

The gOAL: TO BRIDGE THE GAP BETWEEN BUSINESS OWNERS AND THEIR INVENTORY THROUGH A WEB-BASED MANAGEMENT SYSTEM.

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Main Objective

Main objective: To develop a web-based inventory and sales management system tailored to FRACA SERVCOM's furniture and electronics retail operations.

Specific Objective

Build a web-based inventory management system that:

- Implement role-based access for Admin, Manager, and Staff.
- Design a real-time stock tracking engine with automatic low-stock alerts.
- Ensure data integrity using a normalized MySQL database (3NF).
- Professionalization: Replace manual Excel/paper methods with a digital system.
- Economic empowerment: Reduce stockouts on high-demand electronics and reduce overstocking on furniture – directly saving FRACA SERVCOM money.
- Scalability: Provide a foundation that can later support multiple branches or warehouses.

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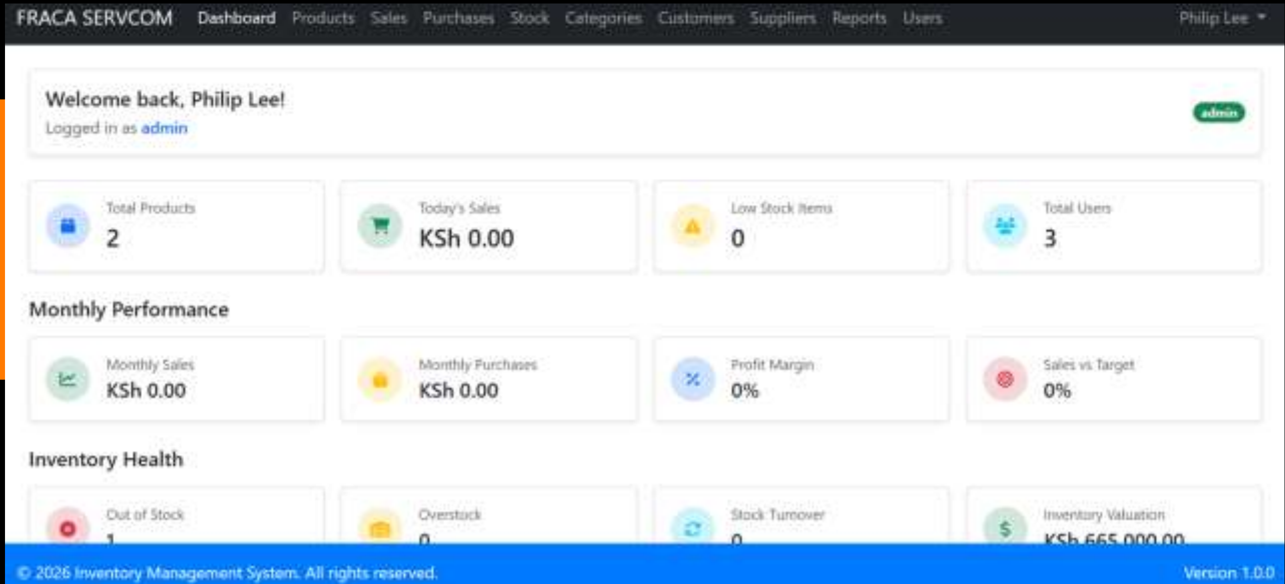
Significance

- First custom inventory system for a Kenyan furniture+hardware+electronics retailer that combines most product types in one unified platform.
- Reduces stockouts (electronics) and overstocking (furniture) through real-time alerts.
- Provides a full audit trail for every stock movement – essential for accountability in a retail business.
- Serves as a digital transformation case study for similar mixed-product retailers in Kenya.



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Literature

- **Existing platforms before this project:** Excel, paper books, generic POS systems (e.g., Loyverse, Odoo) – but they were either too expensive, not tailored to local workflows, or lacked role-based dashboards with hard-delete verification.

- **The gap**: No system offered a simple, affordable, role-based solution with real-time stock alerts AND an audit trail, specifically for a Kenyan furniture+electronics shop.

- **FRACA SERVCOM system advantage**: Built from scratch using Laravel and MySQL, with custom role redirection, admin verification for manager hard-deletes, and full stock history – all localized to FRACA SERVCOM's exact needs.

Methodology

- **Approach**: Iterative Software Development Life Cycle (SDLC) – we built the system in phases.
- **Phase 1**: Backend (Laravel) – Database design (13 tables), models, controllers, REST API, authentication, role middleware.
- **Phase 2**: **Frontend (Bootstrap + Vanilla JS)** – Dynamic dashboards, SPA-style interactions, form validation.
- **Phase 3: Integration & Testing** – Unit tests, feature tests, manual testing with real FRACA SERVCOM data.

TECH STACK:

- **Backend:** Laravel 11 (PHP) - REST API
- **Frontend:** Bootstrap 5 + Vanilla JS (SPA-style, no page reloads)
- **Database:** MySQL (fraca_inventory)
- **Authentication:** Laravel Session Auth + CSRF protection
- **Hosting:** Local (XAMPP) — deployable to any PHP host

RESULTS...

- **Personalized dashboard** – shows logged-in user's name, role badge, and key metrics (total products, low-stock count, today's sales).
- **Product management** – add/edit/delete products with SKU, category, supplier, cost price, selling price, reorder level.
- **Sales recording** – select customer, add products, validate stock, automatically deduct quantity, generate invoice.
- **Purchase orders** – receive stock from suppliers, automatically increase product quantities.
- **Low-stock alerts panel** – real-time list of products with stock ≤ reorder level; dismiss alerts.
- **Stock adjustment** – manual correction with reason and audit trail.
- **Reports** – sales report, purchase report, profit/loss, stock movement history, inventory valuation.
- **Role-based UI** – Admin sees user management; Manager sees everything except hard deletes (requires admin password); Staff only sees sales/purchases.

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ACHIEVED OBJECTIVES & CONCLUSION…

- **Summary: Successfully implemented a fully functional web-based inventory management system for FRACA SERVCOM that meets all SRS/SDS requirements.**

All specific goals achieved:

- Role-based access (Admin, Manager, Staff) – working.
- Real-time stock tracking and low-stock alerts – working.
- Data integrity with full audit trail – working.
- Professionalization – FRACA SERVCOM has replaced Excel/paper.
- Economic empowerment – stockouts reduced, overstocking minimized.
- Scalable foundation – ready for multi-branch or mobile app.

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THANK YOU

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